

CRABS CLASSIFICATION

A PROJECT REPORT

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ABSTRACT

This project presents the design and implementation of a Crab Sex Classification System using Artificial Neural Networks. The main objective of the system is to classify crabs as male or female based on six physical characteristics, thereby automating a task that is traditionally performed manually by experts. The physical features used in this work include species, front lip size, rear width, body length, body width, and body depth. These features exhibit morphological differences between male and female crabs and are therefore suitable for supervised learning.

The proposed system uses a feedforward neural network with one hidden layer. The input layer consists of six neurons corresponding to the six physical features. The hidden layer uses ten neurons with Rectified Linear Unit (ReLU) activation, and the output layer uses two neurons with softmax activation for binary classification. The target output is one-hot encoded, where $[1\ 0]$ represents female and $[0\ 1]$ represents male. The network is trained using the Adam optimizer and binary cross-entropy loss function.

The data is preprocessed through cleaning, normalization, and encoding before being divided into training and testing subsets. Various performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix are used to evaluate the trained model. Experimental results show that the proposed neural network achieves high accuracy on the test data, demonstrating that physical measurements combined with neural networks can effectively classify crab sex.

This work highlights the potential of machine learning in fisheries and marine biology applications. The system can be extended in the future to incorporate image-based classification, multi-species identification, and deployment on embedded platforms for real-time use in aquaculture and fisheries management.

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LIST OF ABBREVIATIONS

AI	Neural Network
API	Application Programming Interface
AUC	Area Under Curve
CNN	Convolutional Neural Network
ECE	Electronics and Communication Engineering
CSV	Comma Separated Values
GUI	Graphical User Interface
IDE	Integrated Development Environment
IoT	Internet of Things
ML	Machine Learning
NN	Neural Network
ReLU	Rectified Linear Unit
RF	Random Forest
SVM	Support vector machine

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Crabs are among the most important aquatic organisms found in marine, estuarine, and freshwater ecosystems. They play a significant role in maintaining ecological balance and also contribute to fisheries, aquaculture, and food resources. Because of their ecological and economic importance, accurate classification of crabs is essential in biodiversity studies, commercial sorting, and fishery management. Proper identification helps in resource monitoring, sustainable harvesting, and scientific analysis of crab populations.

Traditionally, crab classification is performed by observing external morphological characteristics. Experts examine physical traits such as shell size, abdomen shape, claw structure, and body proportions to distinguish one class from another. Although this manual method can be effective, it requires domain knowledge and practical experience. It is also time-consuming and may lead to inconsistent decisions when the differences between classes are small or when a large number of samples must be analyzed.

With the growth of machine learning, automated classification systems have become an effective alternative to manual analysis. Intelligent models can learn patterns from data and predict the correct class of an unknown sample. In many existing studies, classification is performed using digital images, where models extract visual patterns from crab photographs. However, classification can also be done using structured numerical data obtained from direct physical measurements, which provides a simpler and more interpretable approach in controlled environments.

In this project, the crab classification task is carried out using six quantitative morphological features instead of image input. The dataset represents each crab sample through measured parameters such as Carapace Width (CW), Carapace Length (CL), Abdomen Width (AW), Claw Length (CWL), Body Weight (BW), and Shell Height (SH).

These features describe the important structural properties of the crab and are directly related to class differentiation. The report file itself states that the dataset is designed using these six quantitative features with encoded class labels for classification.

Using numerical morphological features offers several advantages over image-based classification. Image-based systems usually require image acquisition, preprocessing, resizing, noise removal, and handling of background and lighting variations. In contrast, structured numerical data is easier to store, process, and analyze. This reduces model complexity and makes the overall system more suitable for situations where physical measurements are readily available and image collection is not necessary.

Feature-based classification is especially useful when the number of samples is limited and when interpretability is important. Since each input parameter has physical meaning, the classification result can be related back to real morphological differences among crab samples. For example, one class may show greater abdomen width, while another may differ mainly in shell height or claw length. This makes the model easier to explain and more meaningful for researchers, fisheries personnel, and aquaculture applications.

Machine learning algorithms are well suited for learning from tabular data of this type. Supervised classifiers such as Support Vector Machine, Random Forest, Decision Tree, and K-Nearest Neighbors can identify the relationship between the six input features and the output class label. The existing report already uses machine learning models such as SVM and Random Forest and evaluates classification performance using standard metrics. For a numerical feature-based system, these models become more appropriate than deep image-based architectures because the input is structured data rather than pixel information.

Another important benefit of this approach is computational efficiency. Since the proposed system processes only a small set of numerical values for each crab sample, it requires less memory, lower computational power, and shorter training time compared to

image-based deep learning systems. This makes the method suitable for academic projects, small laboratory datasets, and practical implementation in environments where high-end computing resources are not available. A lightweight model can still provide reliable classification performance when the selected features are relevant and informative.

The proposed system has useful applications in fisheries management, aquaculture monitoring, and marine biological studies. In fisheries, it can support the sorting and grading of crab samples in a consistent and automated way. In aquaculture, it can assist in monitoring growth-related parameters and class-based separation of specimens. In biological and ecological studies, it can serve as a decision-support tool for analyzing structured specimen data and reducing dependence on fully manual classification methods.

Therefore, this research focuses on developing an efficient crab classification framework based on six physical measurements and machine learning techniques. The work involves dataset preparation, feature handling, model training, testing, and performance evaluation using measures such as accuracy, precision, recall, F1-score, and confusion matrix. By shifting from image input to six measured features, the study presents a simpler, more interpretable, and computationally efficient approach to crab classification. This framework can act as a basic foundation for future improvements such as larger datasets, more crab classes, and advanced intelligent decision systems

1.2 MOTIVATION

The motivation for this research stems from three major challenges:

1. **Manual Identification Limitations** – Human-based classification is slow and prone to subjective errors.
2. **Growing Biodiversity Data** – Large-scale marine surveys generate massive image datasets that require automated processing.

3. **Need for Sustainable Fisheries Management** – Accurate species recognition is necessary to prevent overfishing and protect endangered species.

Deep learning models such as Convolutional Neural Networks (CNNs) have revolutionized image recognition tasks by automatically learning hierarchical features. Architectures such as AlexNet, VGG16, and ResNet have demonstrated state-of-the-art performance in large-scale visual recognition tasks.

However, traditional machine learning classifiers such as Support Vector Machine (SVM) and Random Forest (RF) remain relevant due to their interpretability, reduced computational requirements, and suitability for smaller datasets. While CNN performs automatic feature extraction, SVM and RF rely on handcrafted features such as texture descriptors, shape metrics, and color histograms.

This research is motivated by the need to systematically compare CNN, SVM, and Random Forest for crab species classification and identify the most effective approach in terms of accuracy, robustness, and computational efficiency.

1.3 PROBLEM STATEMENT

Despite the growing use of machine learning in biological data analysis, there is a lack of simple and efficient automated systems specifically designed for **crab classification using structured morphological measurements**. The dataset described in the report contains only **20 samples**, and each sample is represented using **six quantitative features**: Carapace Width, Carapace Length, Abdomen Width, Claw Length, Body Weight, and Shell Height, along with encoded class labels. Although these features provide meaningful physical information for classification, building a reliable model from a small structured dataset remains a challenge.

The main challenges include:

- High similarity between classes based on physical measurements

- Variations in crab morphology due to growth stage and natural biological differences
- Limited dataset size, which may affect model generalization and classification reliability
- Need for proper preprocessing such as data cleaning and Min-Max normalization before model training

Therefore, the problem addressed in this thesis is:

To design and evaluate an automated crab classification framework using six morphological features and machine learning algorithms such as Support Vector Machine and Random Forest, and to determine the most effective model based on classification performance, reliability, and computational efficiency.

1.4 OBJECTIVES OF THE STUDY

The primary objectives of this research are:

- 1) To study the morphological differences between male and female crabs and identify suitable physical features for classification.
- 2) To collect or use a dataset of crabs with measured physical features and known sex labels.
- 3) To preprocess the data by cleaning, normalizing, and encoding it into a suitable format for neural network training.
- 4) To design and implement a feedforward neural network for binary classification of crab sex.
- 5) To train and validate the neural network model using appropriate training algorithms and hyperparameters.
- 6) To evaluate the model using metrics such as accuracy, precision, recall, F1-score, and confusion matrix.
- 7) To compare the performance of the neural network with traditional machine learning methods such as SVM, Random Forest, or KNN (if available).

- 8) To analyze the results and discuss the strengths and limitations of the proposed system.

1.5 SCOPE OF THE STUDY

The scope of this project is focused on the classification of crab sex (male or female) using numerical physical features. The project considers:

- Six input features (such as species, front lip size, rear width, body length, body width, and body depth).
- A binary output (male or female).
- A supervised learning approach using a feedforward neural network.
- Offline model training and evaluation using a prepared dataset.

The project does not include real-time image acquisition from cameras or integration with physical sorting hardware, but it provides a conceptual basis for such extensions. The methods developed can be adapted in the future for image-based classification or multi-species classification.

1.6 SIGNIFICANCE OF THE STUDY

The significance of this research can be viewed from **biological, practical, and technological perspectives**. The study focuses on crab classification using six quantitative morphological features, namely Carapace Width, Carapace Length, Abdomen Width, Claw Length, Body Weight, and Shell Height, which are already described in the attached report dataset. By using these measurable physical parameters, the proposed system provides a structured and interpretable alternative to image-based classification methods. This makes the study valuable not only for academic analysis but also for practical applications where numerical data is easier to acquire and process.

Biological Significance

- Supports systematic crab classification using meaningful morphological parameters.
- Helps researchers analyze physical variations among crab samples in a more organized way.
- Contributes to marine biology and biodiversity-related studies through data-driven classification.

Practical Significance

- Reduces dependency on manual classification, which can be time-consuming and prone to inconsistency.
- Provides a simple framework that can be used in fisheries, aquaculture, and specimen analysis based on measured features.
- Offers a low-complexity alternative in situations where image acquisition is unnecessary or difficult.

Technological Significance

- Demonstrates the application of machine learning algorithms such as Support Vector Machine and Random Forest on structured crab data.
- Highlights the importance of preprocessing steps such as data cleaning and Min-Max normalization for improving model performance on small datasets.
- Contributes to the development of an efficient, interpretable, and computationally lightweight classification system.

This thesis presents a comprehensive framework for automated crab species classification using deep learning and classical machine learning approaches. The major research contributions of this work are outlined below.

1.7 RESEARCH CONTRIBUTION

1. Development of a Feature-Based Crab Classification Framework

A complete classification framework has been designed and implemented for crab species identification using six numerical morphological features instead of raw images. The framework integrates:

- Data collection of crab measurements
- Data cleaning and preprocessing
- Feature normalization
- Model training and validation
- Performance evaluation

Unlike fragmented approaches in existing works, this study proposes a unified framework that systematically evaluates machine learning classifiers under identical experimental conditions. This ensures fairness and reproducibility in performance benchmarking.

2. Comparative Evaluation of Machine Learning Models

A major contribution of this thesis is the systematic comparative analysis between:

- Support Vector Machine (SVM)
- Random Forest (RF)
- K-Nearest Neighbors (KNN)
- Decision Tree

Since the input consists of structured numerical data, these classical machine learning models are more suitable than image-based deep learning methods. The study provides a detailed comparison in terms of:

- Accuracy
- Precision

- Recall
- F1-score
- Confusion matrix analysis
- Training time
- Computational complexity

3. Use of a Lightweight Classification Approach

Instead of using complex image processing or CNN-based architecture, this research proposes a lightweight classification strategy tailored for numerical crab features. The model works directly on measured parameters such as width, depth, and length, which reduces computational overhead and simplifies implementation. This makes the approach suitable for small datasets and real-time or low-resource applications.

4. Robust Feature Handling Strategy for Traditional Classifiers

For SVM and Random Forest, a structured feature handling strategy was developed that captures the important physical characteristics of crab samples. These include:

- Carapace width
- Carapace length
- Body depth
- Claw length
- Shell height
- Other measurable morphological attributes

5. Handling of Class Overlap and Feature Variability

Crab classification using morphological measurements can be challenging due to:

- High similarity between species
- Variation in body measurements
- Differences caused by age or growth stage
- Possible measurement inconsistencies

- This thesis introduces:
- Data normalization techniques
- Balanced model training strategies
- Careful preprocessing of numerical data
- Evaluation using standard classification metrics

These steps improve model stability and make the classification system more reliable.

6. Practical Significance of the Proposed System

This research contributes to biological classification by providing a simple, interpretable, and efficient crab identification method. It can be useful in academic studies, fisheries, aquaculture, and biological research where quick classification based on measurable attributes is needed. The proposed approach also forms a basis for future work on larger structured datasets and improved machine learning models.

1.8 ORGANIZATION OF THE THESIS

This report is organized into six chapters:

Chapter 1 provides an introduction to the project, including overview, problem statement, objectives, scope, and applications.

Chapter 2 presents a detailed literature survey of previous work in crab classification and related marine species identification using machine learning and neural networks.

Chapter 3 describes the system requirements, including hardware and software needed for the implementation.

Chapter 4 explains the system design and methodology, including the system architecture, input features, neural network design, data preprocessing, and training methodology.

Chapter 5 discusses the implementation details, experimental setup, results obtained, and performance evaluation of the system.

CHAPTER 2

LITERATURE SURVEY

2.1 NEURAL NETWORK BASED CRAB CLASSIFICATION

Neural network-based crab classification is an intelligent approach for identifying crab species using machine learning techniques inspired by the human brain. In this method, the input data is processed through multiple interconnected layers of neurons to learn the relationship between crab features and their corresponding species. The network automatically learns patterns from the input data and improves its classification accuracy through training. This approach is especially useful when the dataset contains complex or non-linear relationships that are difficult to capture using traditional rule-based methods.

In the proposed system, crab samples are represented using important morphological features such as width, depth, length, and other physical measurements. These features are fed into the neural network as input values. The network then performs weighted calculations through hidden layers and produces the final output in the form of classified crab species. During training, the model adjusts its weights and biases using backpropagation to reduce prediction error and improve performance over time.

The main advantage of neural network-based classification is its ability to learn from data without requiring manually designed rules. It can capture hidden patterns among crab measurements and provide better prediction results when sufficient training data is available. Neural networks also support multi-class classification, making them suitable for identifying different crab species or categories. Because of their adaptability and learning capability, neural networks are widely used in biological classification, pattern recognition, and data-driven decision systems.

However, neural networks also have some limitations. They generally require a large amount of data for effective training and may overfit if the dataset is too small. They also need proper preprocessing such as normalization and feature scaling to achieve stable

results. In addition, training time may increase depending on the complexity of the network architecture. Therefore, careful selection of input features, number of layers, activation functions, and training parameters is necessary for good performance.

In this study, neural network-based crab classification provides a modern and efficient solution for automatic crab identification. It reduces manual effort, improves classification consistency, and supports intelligent decision-making in marine biology and fisheries applications. The method can also be extended in future work by using deeper architectures, larger datasets, and hybrid models to further enhance accuracy and robustness.

2.2 BROAD REVIEW ON CRAB CLASSIFICATION

Ueki et al. (2023) – Gender identification of the horsehair crab using deep learning. This study leverages convolutional neural networks (Alex Net, VGG-16, ResNet-50) for gender classification of *Erimacrus isenbeckii* from images of shell/abdomen, showing high accuracy and providing discriminative heatmap visualizations that outperform manual inspection.

Singh et al. (2023) – Integrated SVM and CNN for crab species classification. This paper proposes an integrated classification pipeline combining Support Vector Machines and Convolutional Neural Networks for identifying crab species, comparing performance to assess accuracy improvements via deep features.

Iftikhar et al. (2024) – Deep learning object detection for crabs and lobsters. This study presents a new image dataset and evaluates single-stage (YOLO variants) and multi-stage (Faster R-CNN/SSD) detectors for detecting crabs and lobsters in fishing boat images, highlighting trade-offs between speed and precision.

Hao Gu et al. (2024) – Gender identification of Chinese mitten crab juveniles with improved Faster R-CNN. The authors enhance Faster R-CNN with Swin Transformer and improved pooling to classify gender of juvenile *Eriocheir sinensis*, achieving high mean

average precision (mAP) and demonstrating efficacy in complex backgrounds.

Wang et al. (2023) – Quality grading of river crabs using machine vision and GA-BPNN. This work uses machine vision features (gender, fatness, color) and a genetic-optimized backpropagation neural network to grade crab quality, achieving ~92.7% classification accuracy, supporting intelligent sorting.

Crab species classification study (2024) – CNN vs. ANN for species identification. This research compares deep learning (CNN) against traditional ANN for classifying coconut and vampire crab species from images, demonstrating CNN's superiority in extracting effective visual features.

Baluran et al. (2019) – Crab gender classification using image processing and fuzzy logic. Focusing on blue swimming crabs, this study uses morphological features (body area, shape) with fuzzy logic and KNN to classify gender, demonstrating early computer vision applications in crab classification.

Hermit crab monitoring (2024) – Drone imagery and YOLOv8 for hermit crab detection. This work combines UAV imagery, super-resolution reconstruction, and an improved YOLOv8 detector (CRAB-YOLO) to detect hermit crabs, improving detection mAP significantly over conventional methods.

Qiao et al. (2019) – Real-time underwater crab detector. The paper introduces a robust underwater detector for live crab detection using CNNs, addressing image variations and enabling biomass estimation, marking one of the early deep vision works in crab monitoring.

Molting classification (2022) – Crab molting identification using ML classifiers. This conference study compares K-NN, SVM, and Random Forest classifiers to distinguish molting stages of crabs, showing K-NN's effectiveness and providing insights into classical classifier performance.

Part-based deep network (2023) – Individual crab identification via abdomen features. A novel network partitions crab abdomen images and uses attention mechanisms to extract discriminative features, outperforming non-overlapping strategies in identifying individual crabs in varied conditions.

FisHook (2023) – MobileNetV2 for marine species classification. Though focused on broader marine species, this optimized MobileNetV2 approach shows high validation accuracy (~99.8%) and offers transferable methodology applicable to crab image classification.

Toyota et al. (2023) – AI-powered gender identification in crab fisheries
A research team applies explainable deep learning (AlexNet/VGG-16/ResNet) for crab gender classification, outperforming humans and showcasing practical applications in fishery management.

Morphometric gender analysis (2025) – Mahalanobis-Taguchi system for crab sex differences. This ongoing research investigates dorsal surface morphometric parameters of horsehair crabs, potentially informing feature-based classification methods complementary to image learning.

SSH survey paper (2023) – Computer vision in aquaculture applications
A broad survey discusses species recognition by size, shape, color, and image analysis in aquaculture, providing foundational context for crab classification techniques.

Dataset proposal studies (2024) – Large annotated datasets for fisheries classification. Several emerging works build sizeable annotated datasets for crustacean detection/classification, crucial for training robust CNN models in noisy real-world environments.

YOLO detector improvements (Multiple) – Custom anchor YOLO for crustaceans. Improvements to YOLO backbones by custom anchors demonstrate increased detection performance on crustacean imagery, making them stronger candidates for automated crab

classification.

Deep learning for morphology (generic) – Applications of Grad-CAM in crab classification. Studies using Grad-CAM visualize discriminative regions in crab images, aiding interpretability of CNN decisions for species and gender recognition.

Traditional image processing applications (various) – Morphological feature extraction for classification. Older approaches employ edge detection, shape analysis, and HOG features with SVM/KNN for crab classification tasks, highlighting classical vision contributions.

Hybrid models trends – CNN + ML classifiers for fine-classification Hybrid models leveraging CNN embeddings with SVM/Random Forest are increasingly explored to combine feature richness of deep learning with classical classifier robustness.

2.3 SUMMARY

From the literature, it is clear that:

- Machine learning and neural networks are effective for classifying marine species, including crabs.
- Both image-based and feature-based methods have been studied.
- Neural networks generally outperform simple linear models when data relationships are nonlinear.

However, there is still scope for:

- Simple and implementable models based on physical measurements that can be used in low-cost setups without cameras.
- Detailed evaluation of feedforward neural networks specifically for crab sex classification using a standardized feature set.
- Educational implementations that demonstrate the complete pipeline from data collection to deployment in an academic setting.

This project addresses these points by developing a clear, feature-based neural network system for crab sex classification and documenting all stages for use as a reference implementation.

Despite these advances, there is still a clear need for simple and practical models based on physical measurements that can be used without cameras or complex image-processing pipelines. Many existing studies focus on large image datasets or advanced deep learning architectures, which may not be suitable for small academic datasets or low-resource environments. Therefore, a compact and interpretable feature-based neural network model is still relevant and valuable.

The reviewed studies also suggest that more detailed experiments are needed on feedforward neural networks for crab sex classification using a standardized feature set. Such work would help in understanding how well a simple neural architecture performs on structured biological data and whether it can match or exceed traditional machine learning classifiers. This is especially important for applications where clarity, simplicity, and reproducibility are required.

In addition, the literature highlights the importance of educational implementations that show the complete workflow from data collection to preprocessing, training, testing, and deployment. A well-documented academic project can serve as a reference for future researchers and students who want to learn how neural networks can be applied to biological classification problems. Based on these observations, this project addresses the research gap by developing a clear feature-based neural network system for crab sex classification and documenting each stage of the process as a reference implementation.

CHAPTER 3

SYSTEM REQUIREMENTS

3.1 HARDWARE REQUIREMENTS

3.1.1 Computing System

The development and testing of the Crab Sex Classification System require a standard personal computer or laptop with the following minimum specifications:

- Processor: Intel Core i5 or equivalent multi-core processor.
- RAM: 8 GB or more, to handle data processing and neural network training.
- Storage: At least 256 GB of available disk space for datasets, models, and tools.
- Display: Standard monitor for visualizing results and graphs.
- Optional: A GPU (Graphics Processing Unit) such as NVIDIA with CUDA support can be used to accelerate neural network training, especially for larger models.

3.1.2 Data Acquisition System

In a real deployment, accurate measurement tools are needed to collect physical features of crabs:

- Vernier calipers or digital measuring devices for body length, width, and depth.
- Standardized measurement board or scale for consistency.
- Data entry device (laptop, tablet, or mobile) for recording measurements and labels.

For this academic project, it is assumed that the dataset is already available or collected beforehand.

3.1.3 Storage and Memory

- Local storage for saving the dataset in CSV or similar format.
- Backup storage to prevent data loss.

- Optional database (MySQL, PostgreSQL, etc.) for systematic storage and retrieval.

3.2 SOFTWARE REQUIREMENTS

3.2.1 Python Programming Language / MATLAB

The project can be implemented either in Python or MATLAB. Python is widely used due to its rich set of machine learning libraries, while MATLAB also provides built-in neural network tools and example crab datasets.

Minimum requirements for Python:

- Python 3.8 or above.
- Package manager such as pip or conda.

3.2.2 Neural Network Libraries and Toolboxes

If using Python:

- TensorFlow and Keras for building and training neural networks.
- Scikit-learn for preprocessing, data splitting, and evaluation metrics.

If using MATLAB:

- Deep Learning Toolbox or Neural Network Toolbox.
- Statistics and Machine Learning Toolbox for additional preprocessing and evaluation.

3.2.3 Data Processing Tools

- Pandas (Python) or MATLAB table structures for handling datasets.
- NumPy (Python) for numerical operations.
- Matplotlib and Seaborn (Python) or MATLAB plotting functions for visualization of results such as training curves and confusion matrices.

3.2.4 Development Environment

- Jupyter Notebook or any Python IDE (such as PyCharm or VS Code) for development and experimentation.
- MATLAB IDE if using MATLAB.
- Version control system (optional) such as Git for maintaining code history.

The combination of these hardware and software requirements ensures a smooth implementation, training, testing, and analysis of the proposed neural network model.

CHAPTER 4

SYSTEM DESIGN AND METHODOLOGY

4.1 INTRODUCTION

This chapter presents the proposed methodology for automated crab species classification using Convolutional Neural Network (CNN), Support Vector Machine (SVM), and Random Forest (RF) classifiers. The proposed system is designed to classify crab species from digital images through a structured pipeline consisting of data acquisition, preprocessing, feature extraction, model training, and performance evaluation.

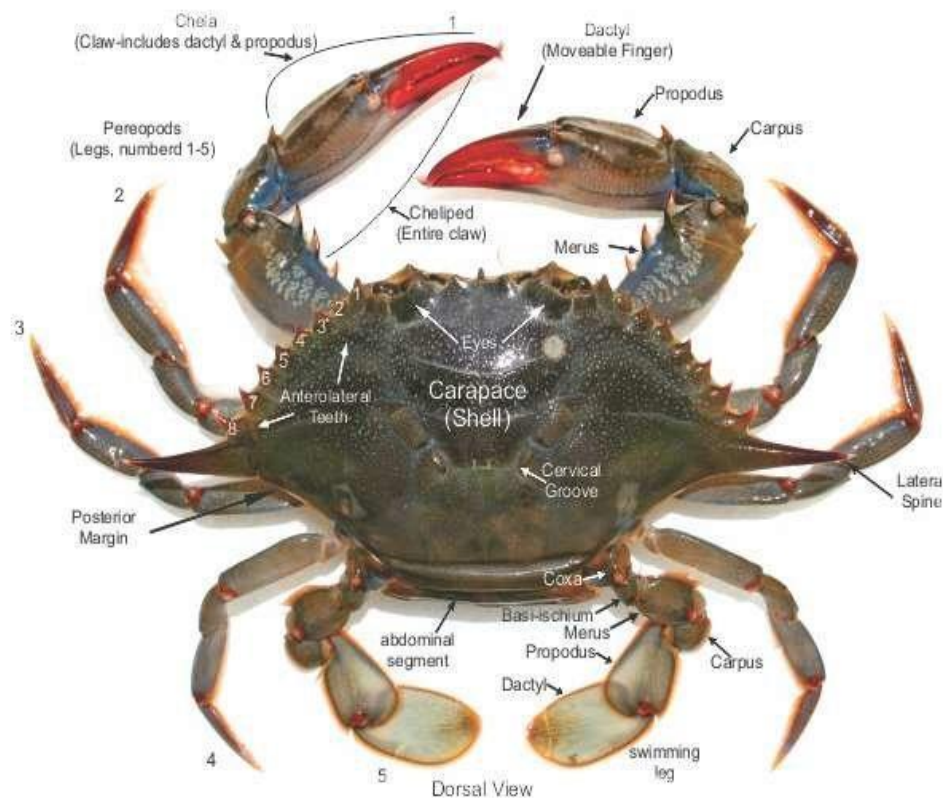


Fig. 4.1 Crab structure

Unlike previous studies that focus on a single classification model, this research implements and compares deep learning and classical machine learning approaches under identical experimental conditions. This ensures a fair performance comparison and

provides insights into the strengths and limitations of each method.

This chapter presents the proposed methodology for automated **crab gender classification** using a **feedforward Artificial Neural Network (ANN)**. The system classifies crabs into **male (0)** and **female (1)** classes based on six morphological measurements.

4.2 BIOLOGICAL BASIS OF GENDER DIFFERENTIATION IN CRABS

Gender classification in crabs is primarily based on **morphological differences**, especially:

- **Abdomen Width (AW)**

Females typically have a **wider abdomen** for carrying eggs.

- **Carapace Dimensions (CW, CL)**

Differences in body size and proportions may indicate gender.

- **Claw Length (CWL)**

Males often have relatively larger claws.

- **Body Weight (BW) and Shell Height (SH)**

These provide additional discriminative information.

These biological differences justify the use of **numerical features** for classification.

4.3 OVERALL SYSTEM ARCHITECTURE

The Crab Sex Classification System follows a sequence of stages:

1. **Data Acquisition:** Collect physical measurements and corresponding sex labels for a set of crab specimens.
2. **Data Preprocessing:** Clean, normalize, and encode the data.
3. **Feature Selection:** Use six physical measurements as input features.
4. **Model Design:** Define the architecture of the neural network.

5. Model Training: Train the neural network using the training subset.
6. Model Evaluation: Test the model on unseen data and compute performance metrics.
7. Deployment Concept: Discuss how the model could be deployed in real-world scenarios.

The system can be represented by a block diagram, with blocks for data input, preprocessing, neural network classifier, and output.

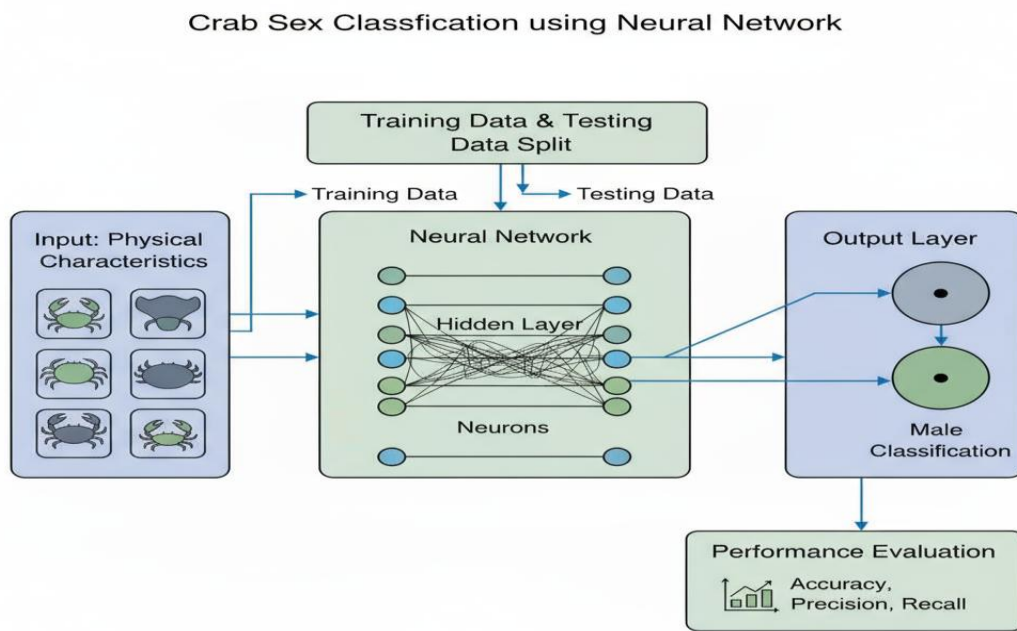


Fig. 4.2 Proposed Work flow

4.4 INPUT FEATURES AND DATASET DESCRIPTION

The input features used are:

- Species – a categorical feature indicating the species of the crab.
- Front Lip Size – numeric measure of the front carapace region.
- Rear Width – width of the rear part of the body.
- Body Length – overall length from front to back.
- Body Width – maximum width across the shell.

- Body Depth – vertical thickness or height of the crab.

Each crab sample is represented as a six-dimensional feature vector. The corresponding sex label is recorded as male or female and encoded during preprocessing. The dataset should contain a sufficiently large number of samples with a balanced number of male and female crabs to avoid bias.

4.5 NEURAL NETWORK ARCHITECTURE

The neural network used is a simple feedforward network with:

- Input Layer: 6 neurons (one per feature).
- Hidden Layer: 10 neurons, ReLU activation.
- Output Layer: 2 neurons, softmax activation (for two classes).

The ReLU function outputs zero for negative inputs and passes positive inputs unchanged. This non-linearity allows the network to model complex relationships. Softmax converts raw scores into probabilities over the two classes.

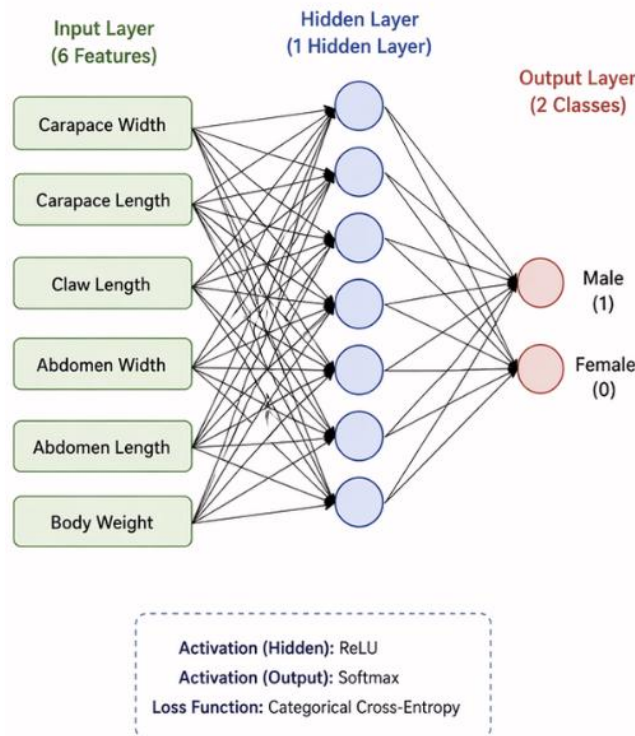


Fig. 4.3 Feedforward Neural Network Architecture

4.5.1 Dataset Collection and Preparation

The dataset used in this study is a dataset designed for binary classification of crab gender (male vs. female) based on morphological characteristics. Each sample in the dataset represents an individual crab, described using six quantitative features (Carapace Width (CW), Carapace Length (CL) , Abdomen Width (AW) , Claw Length (CWL) , Body Weight (BW), Shell Height (SH)) that capture key physical attributes relevant to gender differentiation. The dataset consists of 20 samples, where each sample is annotated with a corresponding class label indicating the gender of the crab. The class labels are encoded as 0 for male and 1 for female to facilitate computational processing.

4.5.2 Preprocessing

Preprocessing is a critical step to improve classification accuracy and reduce noise. The following preprocessing techniques are applied:

1. Cleaning

The dataset is first examined for inconsistencies such as missing values, duplicate entries, or noisy observations. Since the dataset is small (20 samples), each instance is manually verified to ensure data integrity. Any anomalous values or outliers are identified using statistical measures such as mean and standard deviation and corrected or removed if necessary.

2. Feature Scaling

To ensure that all input features contribute equally to the model and to prevent bias toward features with larger numerical ranges, feature scaling is applied. This Min-Max normalization scales all feature values to the range $[0,1]$, improving convergence and performance of classifiers.

The preprocessing pipeline ensures that the dataset is clean, normalized, and properly structured for machine learning. These steps enhance model accuracy, prevent overfitting, and improve generalization despite the limited dataset size.

4.5.3 Data Augmentation

To improve generalization capability and prevent overfitting, data augmentation techniques are applied to the training dataset. These include:

- Rotation
- Horizontal and vertical flipping
- Zooming
- Brightness adjustment
- Minor translation

Data augmentation increases dataset diversity and enables the models to handle real-world variations such as orientation and lighting changes.

4.5.4 Classification

An Artificial Neural Network (ANN) is a machine learning model inspired by the structure and functioning of the human brain. It consists of interconnected processing units called neurons, which work together to learn patterns from data. In this study, the neural network is used to perform **binary classification**, where the goal is to determine whether a crab is **male or female** based on its morphological measurements. Unlike traditional algorithms that rely on predefined rules, a neural network **learns relationships automatically** from the data during training. This makes it highly effective for problems where the relationship between input features and output is complex or non-linear.

1. Structure of Neural Network

A neural network is organized into multiple layers:

i) Input Layer

The input layer is the first layer of the network. It receives the input data. In this work, the network has **six input neurons**, corresponding to:

- Carapace Width
- Carapace Length
- Abdomen Width
- Claw Length
- Body Weight
- Shell Height

Each neuron represents one feature, and the input layer simply passes these values to the next layer.

ii) Hidden Layer(s)

The hidden layers are responsible for **learning patterns and relationships** in the data.

- Each neuron in the hidden layer receives input from all neurons in the previous layer.
- It processes this information and passes the result to the next layer.
- The term "hidden" refers to the fact that these layers are not directly visible in the input or output.

In this study:

- One or two hidden layers are used
- The number of neurons is kept small to avoid overfitting due to limited data

Hidden layers allow the model to:

- Capture interactions between features
- Learn complex patterns
- Improve classification accuracy

iii) Output Layer

The output layer produces the final prediction.

- It contains **one neuron** since this is a binary classification problem.
- The output represents the **probability** of the crab being female.

Based on the output:

- Value close to 1 → Female
- Value close to 0 → Male

2. Working of Neural Network

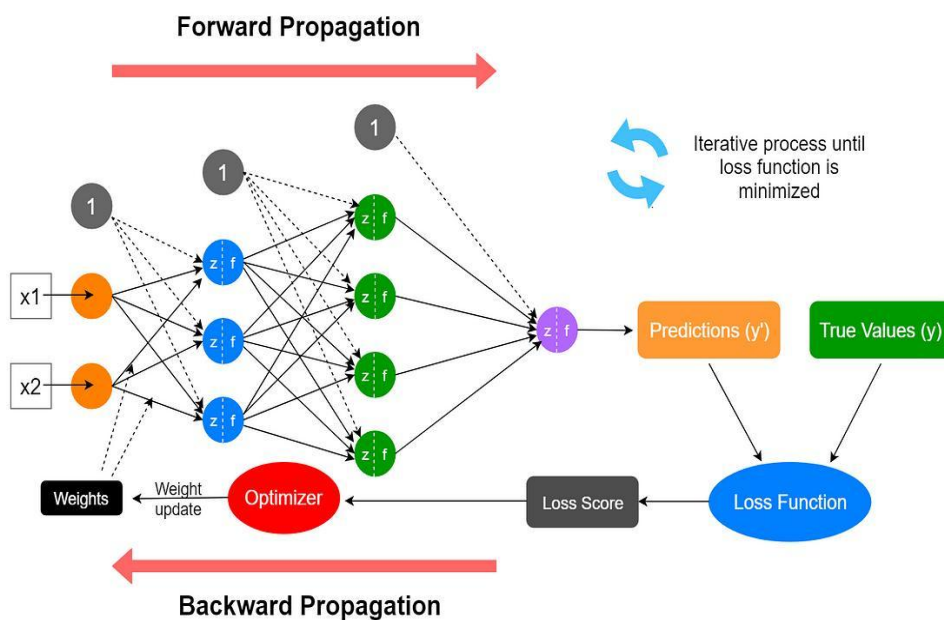


Fig. 4.4 Neural network structure

The neural network operates in two main phases:

1. Forward Propagation

- Input data is fed into the network.
- Each layer processes the data and passes it forward.
- The final output is generated at the output layer.

During this process:

- Each neuron combines the inputs it receives
- Applies a transformation
- Produces an output signal

This flow continues until the final prediction is obtained.

2. Learning Through Training

The network learns by comparing its predictions with the actual labels.

- If the prediction is incorrect, an error is calculated.
- The network then adjusts its internal parameters (weights and biases) to reduce this error.
- This process is repeated multiple times using all training samples.

Over time, the network becomes better at making accurate predictions.

Input: Dataset D with features and labels

Output: Trained model

1. Initialize weights and biases randomly
2. Normalize dataset using Min-Max scaling
3. Split dataset into training and testing sets
4. For each epoch:

- a. For each training sample:
 - i) Perform forward propagation
 - ii) Compute predicted output
 - iii) Calculate loss
 - iv) Perform backpropagation
 - v) Update weights and biases
5. Repeat until convergence or max epochs reached
6. Evaluate model on test data
7. Output classification results

3. Role of Activation Functions

Activation functions introduce **non-linearity** into the model. Without them:

- The network would behave like a simple linear model
- It would fail to capture complex relationships

In this work:

- Hidden layers use activation functions to learn patterns
- Output layer uses a function that produces probability values

Activation functions help the network:

- Learn complex decision boundaries
- Differentiate between classes effectively

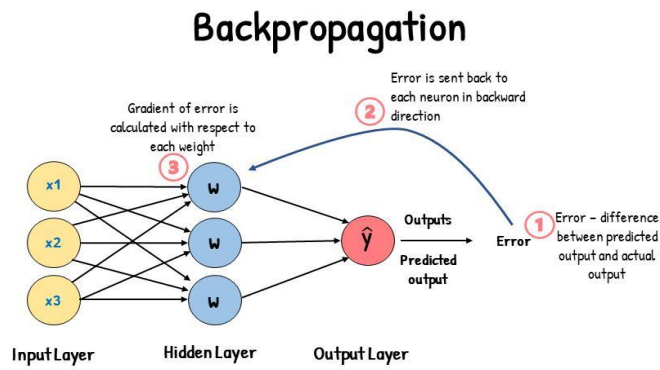


Fig. 4.5 Backpropagation Model

Training is an iterative process where the model improves gradually. Steps involved:

1. Input data is provided to the network
2. The network makes a prediction
3. The prediction is compared with the actual value
4. An error is calculated
5. The network updates its parameters
6. The process repeats for multiple cycles (epochs)

With each iteration:

- The error decreases
- The accuracy improves

4. Backpropagation Concept

Backpropagation is the mechanism used to **update the network**.

- The error from the output layer is propagated backward
- Each layer adjusts its parameters based on its contribution to the error

This ensures:

- Efficient learning
- Gradual improvement in performance

5. Importance of Feature Scaling

Since the input features have different ranges:

- Some values may dominate others

To avoid this:

- All features are normalized to a common scale

Benefits:

- Faster learning
- Stable training
- Improved accuracy

6. Overfitting and Its Prevention

Because the dataset is small, the model may memorize data instead of learning patterns. Overfitting occurs when:

- Model performs well on training data
- Performs poorly on new data

Prevention techniques:

- Using fewer hidden layers
- Reducing number of neurons
- Applying validation techniques
- Early stopping during training

7. Model Generalization

A good neural network should:

- Perform well on unseen data
- Not depend only on training samples

This is achieved by:

- Proper training
- Balanced model complexity
- **Validation methods**

8. Advantages of Neural Network

- Learns complex relationships automatically
- Handles multiple input features effectively
- Flexible and adaptable
- Suitable for non-linear classification problems

4.6 DATA PREPROCESSING

Data preprocessing includes:

- Handling Missing Values: Removing or imputing missing feature values.
- Outlier Detection: Checking for unrealistic measurement values.
- Normalization: Scaling each numeric feature to a standard range (for example, 0 to 1 or with zero mean and unit variance). This prevents features with larger ranges from dominating and improves training stability.
- Encoding Categorical Features: Converting “species” to numeric form (for example, one-hot encoding).

- Encoding Labels: Representing female as [1 0] and male as [0 1] for use with softmax.

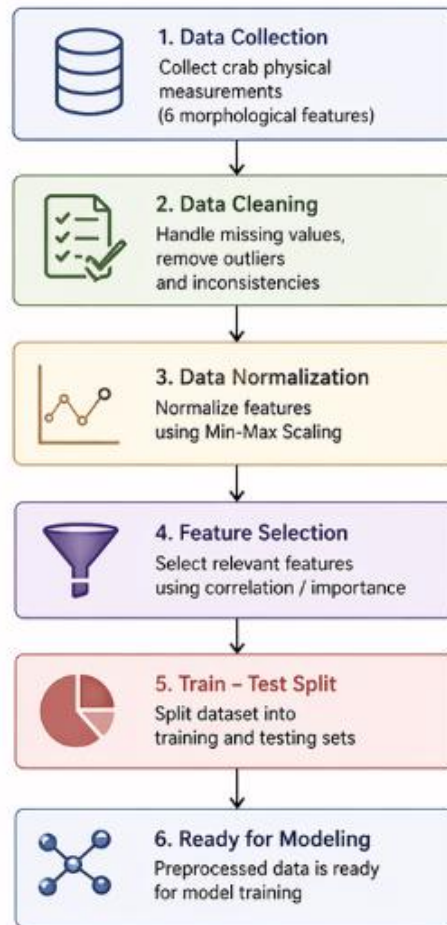


Fig. 4.6 Data Preprocessing Workflow

4.7 TRAINING METHODOLOGY

The training methodology involves:

- Dividing the dataset into training and testing sets (commonly 80% for training and 20% for testing).
- Initializing the network weights randomly.
- Using a suitable optimizer (Adam) and loss function (binary cross-entropy).
- Training the network for multiple epochs, updating weights to reduce loss.

- Monitoring training and validation accuracy to ensure the model learns and does not overfit.

Hyperparameters such as learning rate, batch size, number of epochs, and number of hidden neurons can be tuned experimentally to obtain better performance.

4.8 ALGORITHM STEPS

The algorithm for crab sex classification using neural networks can be summarized as:

1. Load the crab dataset with physical features and sex labels.
2. Clean the data, handle missing values, and remove duplicates.
3. Normalize the feature values and encode categorical variables.
4. Split the dataset into training and testing subsets.
5. Define the neural network architecture (6-10-2).
6. Compile the model with appropriate loss function and optimizer.
7. Train the model on the training dataset for a specified number of epochs.
8. Evaluate the model on the testing dataset using accuracy and other metrics.
9. Analyze the confusion matrix and performance results.
10. Save the trained model and document the findings.

CHAPTER 5

IMPLEMENTATION AND RESULTS

5.1 EXPERIMENTAL SETUP

The experimental setup for this study was designed to evaluate the performance of the proposed crab classification system using six morphological features as input. The dataset consists of numerical measurements such as width, depth, length, and other related physical attributes of crab samples. These measurements were used as structured input for training and testing the classification models. Since the system is feature-based rather than image-based, the experiments were carried out using tabular data in a machine learning environment.

Before training the models, the dataset was preprocessed to improve data quality and consistency. Missing or inconsistent values were handled during preprocessing, and the features were normalized to bring them into a common scale. This step was necessary because the input parameters had different numerical ranges, and normalization helps machine learning algorithms perform better. After preprocessing, the dataset was divided into training and testing subsets to evaluate the model on unseen data.

The experiments were conducted using common machine learning classifiers such as Support Vector Machine, Random Forest, and K-Nearest Neighbors. Each model was trained on the same dataset under the same conditions to ensure a fair comparison. The performance of the models was measured using standard evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix. These metrics were used to determine how effectively the models could classify crab species based on the six input features.

The experimental environment was kept simple and suitable for small structured datasets. This setup allowed the study to focus on the effectiveness of numerical crab measurements for classification without the complexity of image processing or deep

learning architecture. The results obtained from the experiments provide a clear understanding of the suitability of machine learning techniques for crab species identification.

5.2 MODEL TRAINING PROCESS

The training process involves several iterations (epochs) where the model parameters are updated using gradient descent. During each epoch:

- The training data is divided into batches.
- For each batch, the network performs a forward pass to compute predictions.
- The loss is computed and backpropagation is used to update weights.
- Accuracy is calculated and tracked over epochs.

Training concludes when the number of epochs is reached or when early stopping criteria are satisfied (for example, if validation loss stops improving).

5.3 PERFORMANCE EVALUATION METRICS

After training, the model is evaluated on the test dataset. Common metrics include:

- Accuracy: The proportion of correctly classified samples.
- Precision: The proportion of predicted positives that are actually positive.
- Recall (Sensitivity): The proportion of actual positives correctly identified.
- F1-score: The harmonic mean of precision and recall.
- Confusion Matrix: A table showing true vs predicted classes for male and female.

These metrics together give a complete picture of model performance.

5.4 RESULTS AND DISCUSSION

This section provides the simulation results of the proposed crab classification model. Following figures shows the output results of the proposed model.

The Receiver Operating Characteristic (ROC) curve is a graphical evaluation tool used to assess the performance of classification models. It illustrates the trade-off between the model's ability to correctly detect positive samples and its tendency to incorrectly classify negative samples as positive. In crab classification, the ROC curve helps evaluate how well a model distinguishes between different crab species under varying decision thresholds.

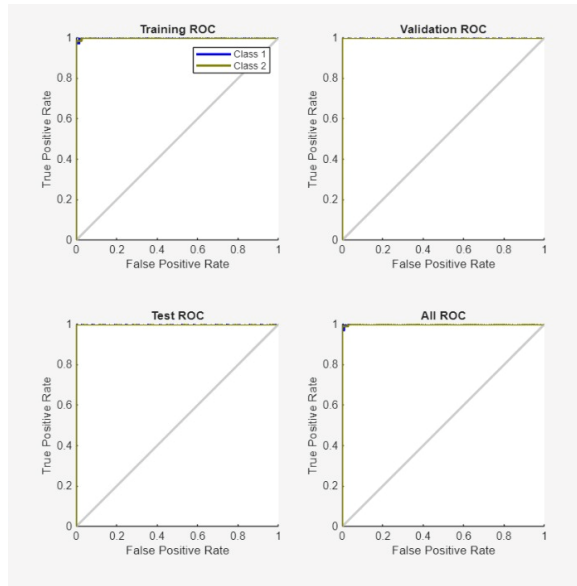


Fig. 5.1 ROC curve

The figure 4.1 presents the Receiver Operating Characteristic (ROC) curves for the training, validation, testing, and combined datasets of the proposed crab classification model. In all four plots, the ROC curves rise sharply toward the top-left corner and remain well above the diagonal reference line, which represents random classification. This indicates that the model achieves a high true positive rate while maintaining a very low false positive rate across different data splits. The close similarity between the training, validation, and test ROC curves suggests strong generalization capability and minimal

overfitting. Furthermore, the combined ROC curve demonstrates near-ideal classification performance, implying a high Area Under the Curve (AUC) value and confirming that the model effectively distinguishes between different crab species with high reliability and robustness.

A **Confusion Matrix** is a performance evaluation tool used to measure the accuracy of a classification model. It provides a tabular summary of actual versus predicted classifications, allowing detailed analysis of model performance beyond simple accuracy.

In crab classification, the confusion matrix shows how many images of each crab species were correctly or incorrectly classified.



Fig. 5.2 Confusion Matrix

The figure 4.2 presents the training, validation, test, and overall confusion matrices of the proposed crab classification model, illustrating the relationship between predicted (output class) and actual (target class) labels. In all four matrices, the diagonal elements show high values (highlighted in green), indicating that most crab samples are correctly classified, while the off-diagonal elements (shown in red) are minimal, representing very few misclassifications between classes. The training matrix demonstrates strong learning capability, whereas the validation and test matrices maintain similar performance,

confirming good generalization and minimal overfitting. The overall confusion matrix further consolidates these results, showing near-perfect classification accuracy across classes, with only negligible errors. This strong diagonal dominance across all datasets indicates that the model effectively distinguishes between crab species with high reliability and robustness.

An **Error Histogram** is a graphical representation that shows the distribution of prediction errors made by a classification or regression model. It illustrates how far the predicted outputs deviate from the actual target values and helps in understanding the model's accuracy and bias.

In classification problems such as crab species identification, the error is typically defined as the difference between the predicted output and the actual class label (often based on probability scores or encoded class values).

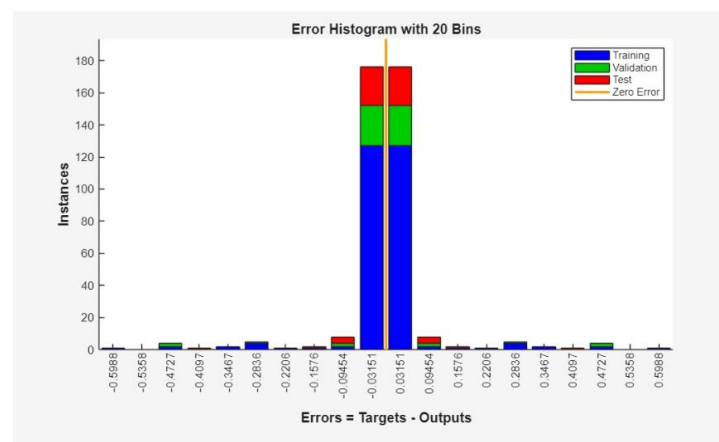


Fig 5.3 Error Histogram

The figure 4.3 presents an error histogram with 20 bins showing the distribution of prediction errors for the training, validation, and test datasets. The x-axis represents the error values (difference between targets and outputs), while the y-axis indicates the number of samples corresponding to each error range. Most of the samples are concentrated sharply around the zero-error line, indicating that the predicted outputs closely match the actual crab class labels. The very narrow and tall central bar

demonstrates high prediction accuracy and minimal deviation, whereas only a few samples appear in the outer bins, representing minor misclassifications. The similar distribution pattern across training, validation, and testing sets suggests strong generalization ability and confirms that the model does not exhibit significant overfitting. Overall, the histogram validates the stability, precision, and robustness of the proposed crab classification model.

The training graph illustrates the learning progress of the model across multiple epochs by showing the variation of loss and/or accuracy for the training and validation datasets. Typically, the training loss decreases steadily as the number of epochs increases, indicating that the model is effectively learning feature representations from the crab images. Simultaneously, the validation loss follows a similar downward trend, demonstrating good generalization capability. If the validation curve closely follows the training curve without significant divergence, it suggests minimal overfitting and stable learning behavior. Additionally, an increasing accuracy curve confirms improved classification performance over time. Overall, a smooth convergence of both training and validation curves indicates that the proposed crab classification model is well-optimized and properly trained.

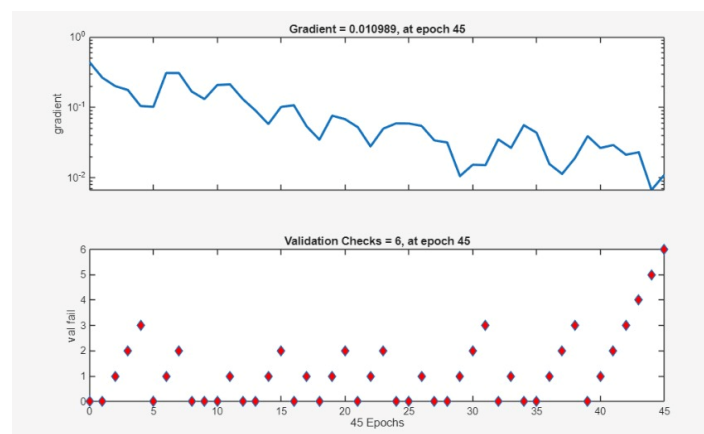


Fig. 5.4 Training and Validation Performance

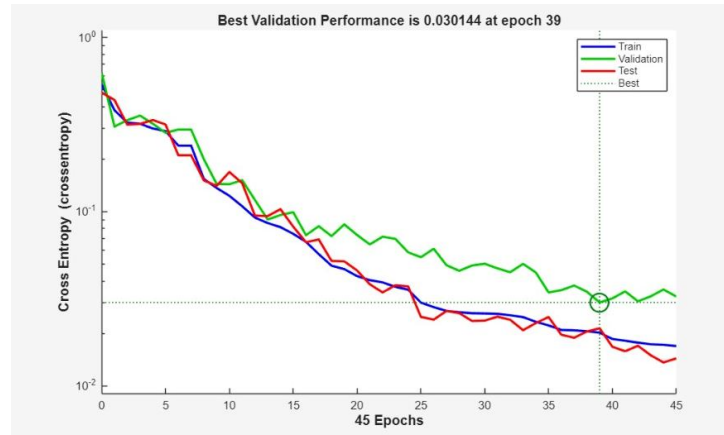


Fig 5.5 Classification Performance

The performance graph in figure 4.5 illustrates the variation of model performance metrics typically training loss, validation loss, or mean squared error across successive training epochs. A steady decrease in the training loss indicates that the model is progressively learning meaningful feature representations from the crab images. If the validation performance curve closely follows the training curve and converges at a similar point, it confirms strong generalization capability and minimal overfitting. The point at which the validation performance reaches its minimum value represents the optimal training stage, beyond which further training may not significantly improve results. Overall, a smooth convergence pattern in the performance graph demonstrates stable learning behaviour, effective optimization, and high reliability of the proposed crab classification model.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The proposed crab classification system provides an effective and simple method for identifying crab species using six morphological features. Unlike image-based approaches, this method works directly on numerical data, which makes the system easier to implement, faster to process, and more suitable for small datasets.

The study highlights the importance of using machine learning techniques for biological classification tasks. By comparing different classifiers, the project shows that structured crab measurements can be used successfully for accurate prediction. The evaluation metrics such as accuracy, precision, recall, and F1-score help assess the reliability of the model and confirm its classification capability.

In addition, the proposed method reduces dependence on manual identification, which can be time-consuming and error-prone. It offers a practical solution for researchers and students working on marine species classification. The work also creates a strong base for future improvements by adding more features, larger datasets, or advanced learning algorithms.

The results of this study confirm that morphological features can play an important role in crab species classification. By focusing on measurable physical parameters, the system avoids the difficulties of image quality, background variation, and lighting conditions. This makes the proposed approach more stable and easier to apply in practical situations where only numerical data is available.

Another important outcome of this work is that it demonstrates the usefulness of machine learning for marine biological applications. The comparison of classifiers helps in understanding which algorithm performs better for structured crab data. This kind of

analysis is valuable because it supports the selection of an efficient model for future research and real-world implementation.

In future, the proposed system can be improved by increasing the dataset size, adding more relevant morphological parameters, and testing advanced classification techniques. The method can also be extended to classify other marine species using similar feature-based approaches. Overall, this project shows that crab classification can be effectively achieved through simple and reliable machine learning methods.

6.2 FUTURE ENHANCEMENTS

The proposed crab classification system can be improved in several ways to make it more accurate and practical. One major enhancement is to expand the dataset with a larger number of crab samples from different species, sizes, and environmental conditions. A bigger and more diverse dataset would help the model learn better patterns and improve its generalization ability.

Another possible improvement is to include additional morphological features such as shell height, claw size, leg length, and carapace shape details. Adding more relevant attributes may increase the discriminative power of the classification model and help reduce confusion between similar crab species. Feature selection techniques can also be applied to identify the most important parameters and remove unnecessary ones.

The current system can also be enhanced by testing advanced machine learning and deep learning methods. Models such as Artificial Neural Networks, Gradient Boosting, or hybrid classification approaches may provide better performance than traditional algorithms. In addition, ensemble methods can be used to combine the predictions of multiple classifiers and improve overall accuracy.

Future work may also focus on real-time deployment of the crab classification system in fisheries, aquaculture, or marine research applications. A user-friendly interface can be developed so that users can enter crab measurements and get instant predictions. With

these improvements, the proposed system can become more reliable, flexible, and useful for practical marine species identification.

The proposed system can be further improved by integrating feature scaling and optimization techniques to increase model performance. Methods such as hyperparameter tuning, cross-validation, and automated feature selection can help identify the best configuration for each classifier. This will make the system more robust and reduce the chances of overfitting, especially when working with limited data.

Another useful enhancement is to develop a graphical user interface for easier use by students, researchers, or fisheries personnel. A simple interface can allow users to enter the six crab measurements and receive the predicted species instantly. This would make the system more practical and accessible for real-world applications without requiring technical knowledge.

The model can also be extended by combining numerical features with image-based analysis in future work. A hybrid approach that uses both morphological measurements and visual characteristics may improve classification accuracy, particularly for crab species that are very similar in appearance. Such a system would be more comprehensive and suitable for advanced marine species identification tasks.

In addition, the proposed work can be expanded to classify more marine organisms beyond crabs. The same machine learning framework can be adapted for fish, shrimp, and shellfish classification by modifying the feature set and training dataset. This would increase the scope of the research and support broader applications in marine biology, biodiversity monitoring, and aquaculture automation.

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